

Adaptive Modulation Switching in 2D Wireless Channels

Shatrajeet Mohapatra^{1*}, Arunangshu Pal^{2#}, Samriddhi Ghosh^{3#}, Ritaja Ghosh^{4#}, Dr. Soumyo Chatterjee⁵

Department of Electronics and Communication Engineering,

Heritage Institute of Technology, Kolkata^{1,2,3,4,5}

Correspondence: Shatrajeet Mohapatra(shatrajeet2005@gmail.com)

#Equal Contribution : The authors contributed equally to the work

Concept Summary—Adaptive modulation improves wireless communication performance by dynamically selecting modulation schemes according to channel conditions. This work presents a MATLAB-based adaptive modulation switching system for a two-dimensional terrestrial wireless channel using BPSK, QPSK, and 16-QAM modulation techniques.

The system utilizes Signal-to-Noise Ratio (SNR) for modulation selection and additionally introduces adaptive threshold-based switching, where threshold values vary according to environmental parameters such as path loss, fading conditions, and receiver mobility.

System performance is analyzed under varying distance, transmit power, carrier frequency, Rician fading, and receiver velocity. Simulation results demonstrate that adaptive modulation maintains reliable communication under poor channel conditions while improving spectral efficiency under favorable conditions. The adaptive threshold mechanism further enhances modulation selection by providing a more realistic and environment-aware switching strategy.

The proposed work highlights the importance of intelligent modulation adaptation in modern wireless communication systems.

Keywords—*Adaptive Modulation, Wireless Communication, BER*

I. ORIGIN OF THE PROPOSAL

Modern wireless communication systems operate under continuously varying channel conditions caused by distance, fading, interference, and user mobility. Conventional communication systems generally employ fixed modulation schemes regardless of channel quality. Although lower-order modulation techniques such as BPSK provide reliable communication under poor channel conditions, they suffer from low spectral efficiency. In contrast, higher-order schemes such as 16-QAM achieve higher data rates but become highly sensitive to noise and fading.

This trade-off between reliability and throughput motivated the development of adaptive modulation techniques. Adaptive modulation dynamically changes the modulation scheme according to channel conditions, thereby maintaining an optimal balance between communication reliability and spectral efficiency.

The proposed work originated from the need to study how multiple environmental and system parameters influence adaptive modulation performance in practical two-dimensional terrestrial wireless channels. Unlike conventional studies that primarily focus only on SNR variation with distance, this work investigates the combined influence of transmit power, path loss exponent, carrier frequency, fading conditions, and receiver mobility.

Additionally, the work introduces an adaptive threshold-based switching mechanism in which SNR thresholds vary according to environmental conditions. This enables more realistic and environment-aware modulation selection compared to conventional fixed-threshold systems.

II. METHODOLOGY

The proposed system models a Single-Input Single-Output (SISO) wireless communication link in a two-dimensional terrestrial environment using MATLAB simulations. Random binary data is generated and transmitted through adaptive modulation schemes including BPSK, QPSK, and 16-QAM.

The wireless channel is modeled using log-distance path loss, Rician fading, and Additive White Gaussian Noise (AWGN). The path loss is calculated using:

$$PL(d) = PL_0 + 10n\log_{10}(d) \quad \dots (1)$$

where PL_0 represents the reference path loss, n is the path loss exponent, and d is the transmitter-receiver distance.

For frequency-dependent analysis, free-space path loss is calculated as:

$$PL = 20\log_{10}(4\pi d/\lambda) \quad \dots (2)$$

where:

$$\lambda = c/f_c \quad \dots (3)$$

Here, λ is the wavelength, c is the speed of light, and f_c is the carrier frequency.

The Signal-to-Noise Ratio (SNR) is obtained as:

$$SNR(dB) = P_t - PL(d) \quad \dots (4)$$

where P_t is the transmit power. The SNR value determines the modulation scheme selected by the adaptive modulation system.

Conventional threshold-based switching is implemented using:

- BPSK for $SNR < 10$ dB
- QPSK for $10 \text{ dB} \leq SNR < 18$ dB
- 16-QAM for $SNR \geq 18$ dB

Additionally, this work proposes adaptive threshold-based switching in which threshold values can vary according to environmental parameters such as path loss exponent, receiver velocity, and fading conditions to achieve more realistic modulation selection

The Rician fading channel is represented as:

$$r(t) = h(t)s(t) + n(t) \quad \dots (5)$$

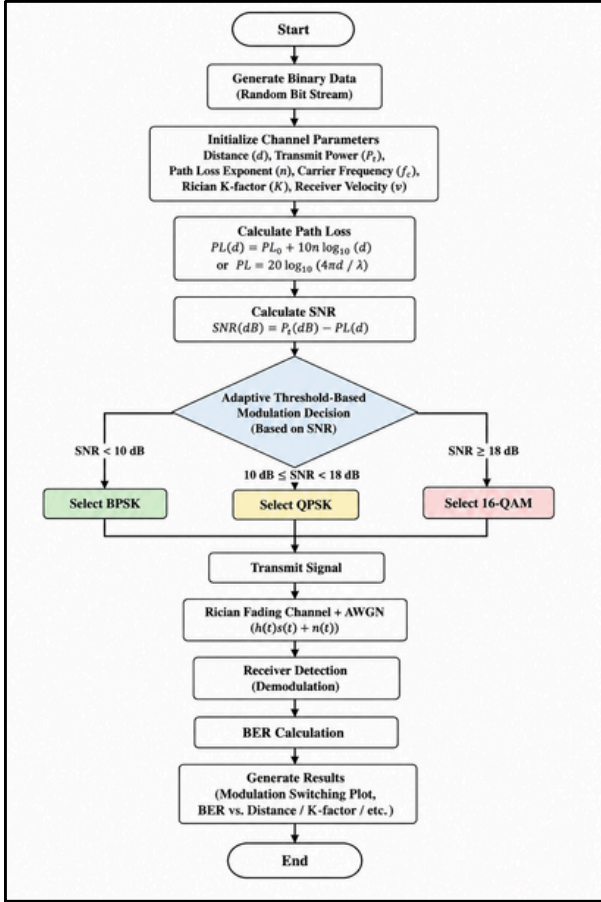
where $h(t)$ represents the fading coefficient and $n(t)$ represents AWGN.

Receiver mobility is incorporated using Doppler frequency:

$$f_d = v/\lambda \quad \dots (6)$$

where v is receiver velocity and λ is the wavelength.

Simulation graphs are generated by varying system parameters including distance, transmit power, path loss exponent, carrier frequency, Rician K-factor, and receiver velocity. Modulation switching behavior and Bit Error Rate (BER) are analyzed to evaluate system reliability and spectral efficiency under different wireless channel conditions.



III. EXPERIMENTAL RESULTS

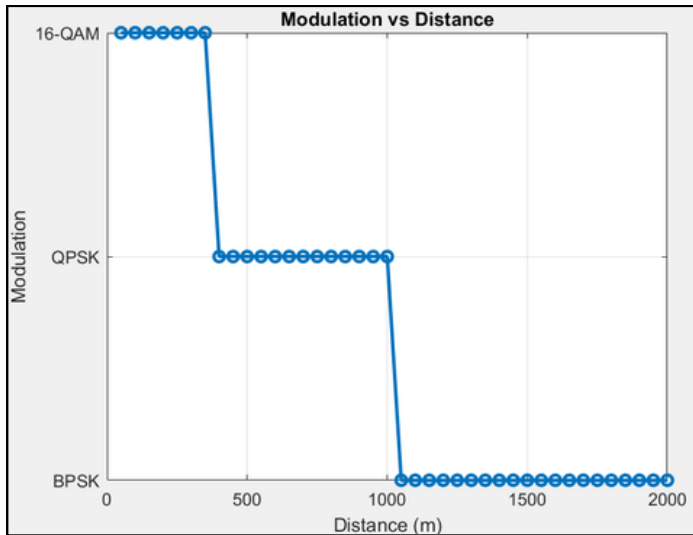


Fig. 1. Modulation switching behavior with increasing distance.

Figure 1 illustrates the adaptive modulation switching behavior with increasing distance. At shorter distances, the Signal-to-Noise Ratio (SNR) remains high, allowing the system to utilize 16-QAM for higher spectral efficiency. As distance increases, path loss causes SNR degradation, forcing the system to switch to QPSK and eventually BPSK to maintain reliable communication.

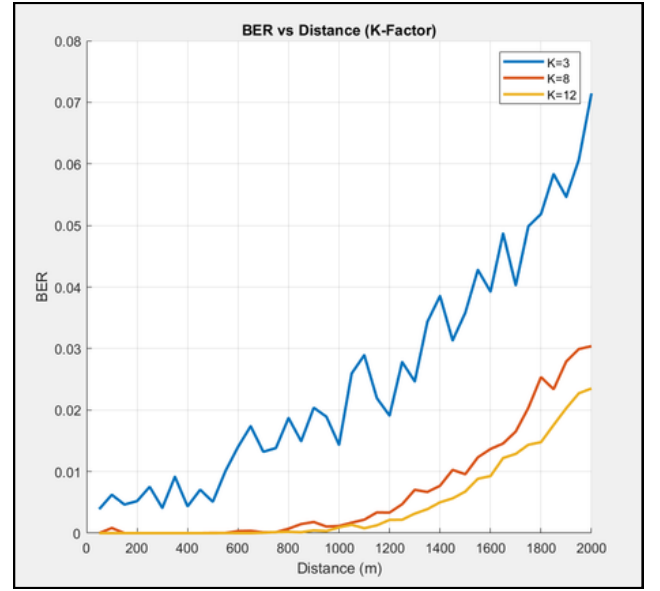


Fig. 2. BER performance under varying Rician K-factor conditions.

Figure 2 shows the Bit Error Rate (BER) performance under different Rician fading conditions. Higher K-factor values result in lower BER due to the presence of stronger Line-of-Sight components, while lower K-factor values increase BER because of stronger multipath fading effects. The results demonstrate the influence of fading conditions on system reliability.

Due to space limitations, only representative modulation switching and BER plots are included in this extended abstract. Additional simulation results were also obtained for varying transmit power, carrier frequency, path loss exponent, receiver velocity, and adaptive threshold-based switching conditions.

The simulation results confirm that adaptive modulation effectively balances reliability and spectral efficiency under varying wireless channel conditions.

IV. INNOVATION AND POTENTIAL APPLICATION

Future work will focus on FPGA-based implementation of the adaptive modulation switching system for real-time wireless communication. The design will support two-dimensional adaptive switching considering spatial and temporal channel variations. Future enhancements may include machine learning-based modulation selection systems and integration with MIMO, 5G, and IoT communication systems.

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